

Executive Summary: This week the team made the most amount of progress and the most critical failure of the project. A major mistake was made in ordering the incorrect microphone and instead of receiving the planned (and originally ordered) I2S digital microphone, we instead received a less powerful analog microphone. However, the team quickly shifted by modifying schematics and software planning such that this should only turn out to be a small set back and not the end of the project. This week we also ran several tests on the GPS Module and found a high accuracy for longitude and latitude. However, we identified significant inconsistencies in altitude reading leading to software based fixes hopefully. In terms of the software alone, this week the team made headway on GPS & time logic by making the timestamps the correct time zones and included cases of leap years. Following the hardware change the audio software was rewritten to process analog voltage instead of digital. Similarly circular buffers were implemented to capture 10-11 ms windows of audio, which is optimal for gunshot durations of 3-5 ms.

Hardware:

Cody C.

Progress:

- 4/8 – Team meeting (~2 Hour)
 - Talked about next steps for project
 - Updates on parts
 - Waiting on I2S microphone
 - Breadboard and protoboard came in and ready to use
 - Talk about plans to build the hardware device before next Wednesday
 - Troubleshooted GPS module

- Test field testing for the GPS
 - Testing the accuracy of the device compared to Google maps, Apply maps, and Galaxy watch (altitude) in reference to website.
 - Altitude fluctuations occur inconsistently, moving even when stationary
 - Longitude and latitude are the most accurate in terms of matching with Google Maps, Apple Maps, Watch, and website.
- 4/9 – Microphone and diode parts arrive
 - Microphone ordered was the wrong version
 - PM checked and the parts are out of stock.
 - Microphone Change from digital microphone to analog microphone
- New microphone
 - Research new maximum dB of the microphone (110 dB)
 - Analog microphone instead of digital microphone
 - Modify old schematics (~20 Minutes)
 - Created new symbol for new microphone (~5 Minutes)
- Solder pins to new microphone (~30 Minutes)

Total Hours Worked: ~ 2 Hours 55 Minutes

Future Work/Plan

- Start PCB prototype in KiCad
- Research how to create a PCB for the prototype to have 3D model to show

- Meet up with Ryan to build whole hardware device

Hardware:

Ryan K.

Progress:

- 4/8 - Team Meeting (~2hrs)
 - Talked about future plans for the project
 - Field tested the GPS module to reveal noticeable bugs and issues early
 - Started construction of complete WAD prototype
- 4/10 - Variety Pack of Inductors Acquired
 - All hardware components are now available and in hands

Total Hours Worked: (~2hrs)

Future Work/Plan:

- Finalize WAD prototype
- Begin testing microphone and overall prototype functionality

Software:

Alex H.

Progress:

- 4/6 (1 h 22 min)
 - Tested GPS functionality in a residential home.

- GPS signal can be received at any part of a residential home. **Therefore, the current signal range of the GPS is any location outside and residential homes. Receiving a signal inside of a commercial building is still not doable.**
- Realized that timestamp conversion from GMT time zone to CST time zone fails when it is during the evening in CST time zone.
- Troubleshooted time zone conversion bugs in software, discovered that software did not accommodate for scenarios when CST and GMT are on different days.
 - Added more sophisticated time zone conversion software that accommodates when CST and GMT are on different days. New time zone conversion software also can detect when the current year is a leap year.
- 4/8 (2 h)
 - Tested the GPS signal range and altitude by walking around the Engineering Quad with the team.
 - Compared GPS latitude and longitude coordinates with GPS coordinates on phone. GPS coordinates work perfectly outside. When the GPS signal is lost inside of a commercial building, the last known latitude and longitude coordinates are preserved until the signal is re-obtained by the module.
 - Compared altitude measurements with altitude measurements on phone. Noticed that altitude measurements fluctuate drastically and inconsistently, especially when someone is standing still. Measurements can fluctuate as small as 0.5 m to as large as 40 m when standing still. The team discussed

a possible averaging function that could be added to the software to regulate the altitude measurement fluctuations.

- Attempted to receive a GPS signal inside of Devon Energy Hall, but it was not possible. Tested GPS signal strength using our phones. Only Cody's phone was able to receive a strong enough signal to track our location inside of Devon Energy Hall.
- 4/9 (15 min)
 - Microphone chips were delivered to the ECE office. Walked with Jonas to obtain the microphone chips. Discovered that the incorrect microphone chip was delivered to our team. The microphone chips received were analog microphones, not I2S-based microphones. With limited time to develop the project, it was decided that the team would have to readjust the design to work with an analog microphone.
 - Soldered pin headers onto a microphone chip for software testing.
- 4/11 (8 h 40 min)
 - Added averaging function to altitude measurements in the software to improve accuracy.
 - Redid audio software to read voltage from an analog microphone instead of digital data from an I2S-based microphone.
 - Tested voltage readings from the analog microphone chip. Voltage readings seemed reasonable based on information about the amplifier gain and DC bias from the chip's datasheet.

- Researched how to convert voltage readings from microphone chip into dB levels (Sound Pressure Level or SPL). Wrote mathematical equations for this conversion.
- Implemented mathematical equations for voltage to dB SPL conversion in the software and tested accuracy of dB SPL level being calculated.
 - Software accurately raises dB SPL level when a loud noise is heard (ex: a loud clap is between 80-90 dB SPL, and software measures that accurately).
 - However, the dB level seems high when only ambient noise can be heard. Typical ambient noise reads between 40-50 dB on a microphone. However, the software calculates between 60-70 dB for ambient noise. This may be due to the microphone having a high noise floor.
- 4/12 (3 h)
 - Developed audio peak detection in the software.
 - If the audio exceeds a certain dB SPL threshold, the software will catch that and indicate that a gunshot may possibly be detected.
 - This threshold can be adjustable in the software.
 - Recalculated size of the main audio buffer. Having an audio buffer that contains the last second of audio, considering that we are sampling audio at 44.1 kHz, is too large for the software to handle. Reduced the audio buffer to contain only the last 10-11 ms of audio, which should be enough space to contain possible gunshot audio (gunshots last for about 3-5 ms).
 - Implemented the main audio buffer in the software as a circular buffer.

Total Hours Worked: 15 h 17 min

Future Work/Plan:

- Continue software development on gunshot detection.
- Determine what the noise floor is for the analog microphone chip by looking through the datasheet and looking online
- Start working on user manual deliverable for the final demo
- Start working on software deliverable for the final demo
- Determine how to collect proper data to show that the Wearable Alert Device can work.

Project Manager:

Jonas W.

Progress:

- 4/8 – Team meeting (~2 Hour)
 - In this team meeting we talked about future steps for the project and I personally handed off the main Protoboard and breadboards for the prototyping.
 - We also walked around the Engineering quad testing the accuracy of the GPS module and found it to be on par if not better than phone for GPS position but we are still working on some of the kinks for the elevation detection of the device.
- 4/9 Start Work on Final Demo Deliverables (~2 hour)
 - I began with writing out the skeleton of the Design Document to help the Hardware group be able to finish their task for this demonstration
 - I started a document for the user manual and wrote some of the introductory details of the manual.

- Finally I began my BOM for the device and made sure that everything we had currently was on there while waiting for the final parts to arrive
- 4/9 Final part pickup and deliver(30 Min)
 - We received our final parts in the mail and after picking them up I handed them off to the team members needing them most. We did find out that the part for the microphone was the wrong part, HOWEVER, the part number listed on digikey was the same part listed on the part parent company because all their devices were labeled the same.
 - Ensured the switched device would be okay and work with all parts and got confirmation that we would still be okay
- 4/10 Final Demo powerpoint creation (30 Min)
 - Created the beginning of the final presentation slides and made sure the number out the exact needed parts
- 4/11 Pick up flash drives for archive of project (30 Min)
 - Drove to store
 - Picked up 2 16 GB that should be able to hold all data from project

Total Hours Worked: 5 hr 30 Min

Future Work/Plan:

For next week I plan on continuing work on the deliverables of the final demo and the final archives of the project. Once we have the software ready for testing on the gunshot detection which could be this week, we plan on making our field test of shooting a gun near the device.

